Questions

Q1.

(a) Which **one** of these statements about alpha radiation is correct?

Put a cross (☒) in the box next to your answer.

(1)

☐ **A** Alpha radiation has no charge.
☐ **B** Alpha radiation is very ionising.
☐ **C** Alpha radiation travels very far in air.
☐ **D** Alpha radiation is an electromagnetic wave.

(b) When an atom emits an alpha particle its nucleus changes.

Which describes the changes in the nucleus?
Put a cross (☒) in the box next to your answer.

(1)

	proton number	mass number
☐ **A**	decreases by 2	decreases by 4
☐ **B**	increases by 2	decreases by 4
☐ **C**	decreases by 2	increases by 4
☐ **D**	increases by 2	increases by 4

Q2.

A teacher sets up an experiment to show some students how far beta particles travel in air.

Figure 10 shows some of the equipment she uses.

(Source: *www.einstein.yu.edu*)

Figure 10

(i) State the scientific name for the radioactivity detector shown in Figure 10.

(1)

.....

The teacher also has:

- a radioactive source that emits only beta particles
- a metre rule.(ii) State **two** precautions the teacher must take to protect herself from the effects of radioactivity.

(2)

1 .....

.....

2 .....

.....

(iii) Describe how the teacher could show how far beta particles travel in air.

(4)

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(Total for question = 7 marks)

Q3.

An isotope of krypton, krypton-89, is produced in a nuclear reactor.
A nucleus of this isotope can be represented as

$$^{89}_{36}Kr$$

Describe the structure of a nucleus of krypton-89.

(4)

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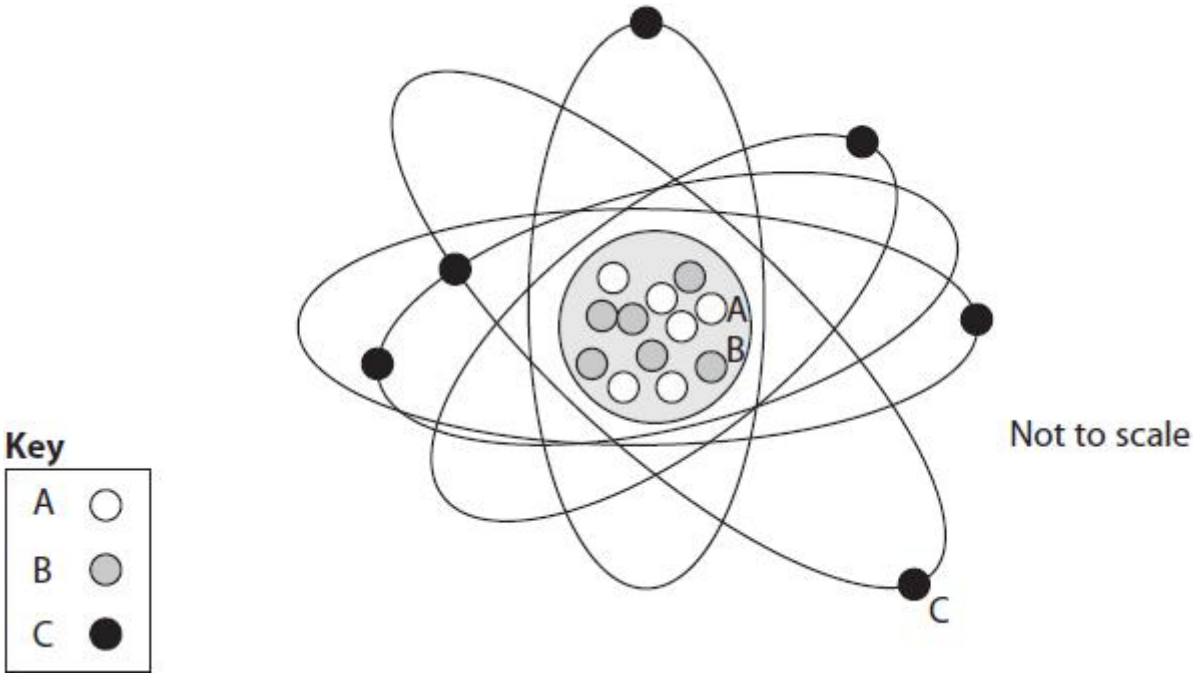
Q4.

The diagram shows an atom of carbon.

A, B and C are three different particles.



(i) Name the three different particles shown.

(3)

A = ...

B = ...

C = ...

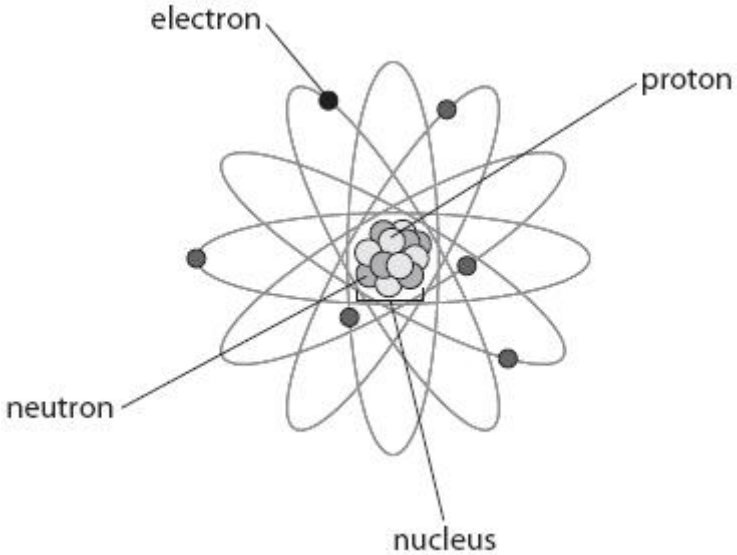
(ii) What is the mass (nucleon) number of this carbon atom?

(1)

.....

Q5.

The diagram shows the structure of an atom.



(a) (i) Complete the sentence by putting a cross (■) in the box next to your answer.
The size of the charge on each electron is

(1)

☒ **A** a third of the charge on the proton

☒ **B** half the charge on the proton

☒ **C** the same as the charge on the proton

☒ **D** twice the charge on the proton

(ii) Complete the sentence by putting a cross (■) in the box next to your answer.
The atomic number of a neutral atom is always the same as the number of

(1)

☒ **A** electrons

☒ **B** electrons and neutrons

☒ **C** protons and neutrons

☒ **D** neutrons

(b) The element radium has a radioactive isotope, radium-226.

This can be written as

82	84	86	90	222	224	228	230

This radioactive isotope emits alpha particles.

The alpha particle has a mass number of 4 and contains two protons.

Using the numbers in the box complete the following sentences.

(i) When an alpha particle is emitted

by

82	84	86	90	222	224	228	230

the mass number becomes .....

(1)

(ii) When an alpha particle is emitted

by

82	84	86	90	222	224	228	230

the atomic number becomes .....

(1)

(c) Describe how the emissions from radioactive substances can be dangerous to living things.

(2)

.....

.....

.....

.....

(d) Explain **one** precaution that is taken in hospitals to limit the risks of exposure to radiation.

(2)

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.....

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.....

(Total for Question = 8 marks)

Q6.

The mass of a proton is 1.6726×10^{-27} kg.
The mass of an electron is 9.1094×10^{-31} kg.

Calculate how many times the mass of a proton is greater than the mass of an electron.

Give your answer to two significant figures.

(3)

..... times

(Total for question = 3 marks)

Q7.

Iodine-131 is a radioactive isotope with a half-life of 8 days.

The activity of a sample of iodine-131 is 480 Bq.

Calculate the activity of the sample after 16 days.

(2)

activity = ... Bq

(Total for question = 2 marks)

Q8.

A sample of a radioactive isotope has a mass of 520 g.
The half-life of the radioactive isotope is 18 days.
(i) Calculate the mass of the original radioactive isotope remaining after 18 days.

(1)

mass after 18 days ..... g

(ii) Calculate the mass of the original radioactive isotope remaining after 54 days.

(2)

mass after 54 days ..... g

(Total for question = 3 marks)

Q9.

Both U-235 and oil can be used as energy sources for generating electricity.

1 kg of natural uranium can result in the generation of 45 000 units of electricity.

1 kg of oil can result in the generation of 5.0 units of electricity.

Calculate the mass of oil needed to generate the same amount of electricity as 1kg of natural uranium.

(2)

mass of oil = ... kg

(Total for question = 2 marks)

Q10.

The isotope fluorine-18 is used in PET scans, which help to detect cancers.

Figure 20 shows a radioactive decay curve for a sample of fluorine-18.

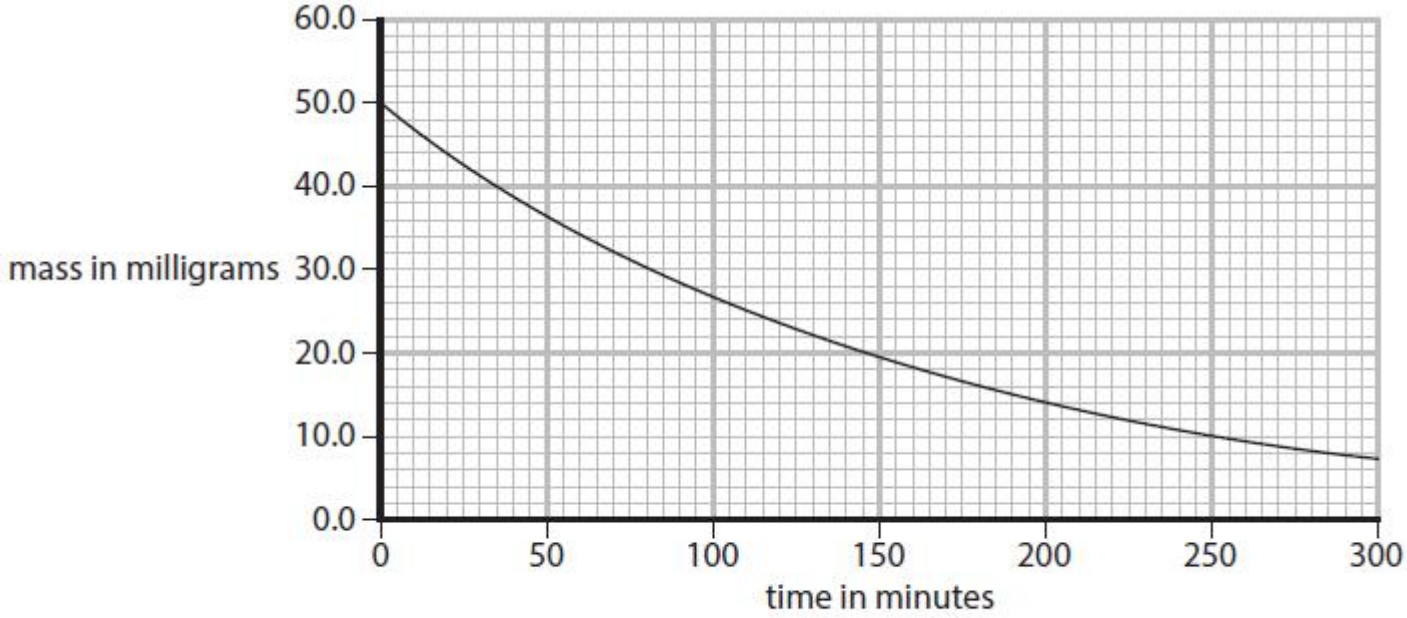

Figure 20

(i) Use the graph to calculate the percentage of fluorine-18 remaining after 150 minutes.

(2)

percentage of fluorine-18 remaining after 150 minutes = ... %

(ii) Fluorine-18 releases positrons.

Describe what happens when positrons meet electrons.

(2)

.....

.....

.....

.....

(Total for question = 4 marks)

Mark Scheme

Q1.

Question Number	Answer	Acceptable answers	Mark
(a)	**B** It is very ionising		(1)

Question Number	Answer	Acceptable answers	Mark
(b)	**A** decreases by 2 decreases by 4		(1)

Q2.

Question number	Answer	Additional guidance	Mark
(i)	Geiger-Müller tube	accept Geiger (counter) geiger (counter) GM (tube) gm(tube) accept any recognisable (phonetic) spelling	(1)

Question number	Answer	Additional guidance	Mark
(ii)	any **two** from: keep a safe distance (1) point the source away from people (1) handle the source with tongs/at a distance (1) limit exposure time/return source to store (asap) (1) use shielding (1) use of gloves (1) use of mask (1) protective clothing (1) wear a film badge/monitor (1)	 use of screen Do not credit goggles	**(2)**

Question number	Answer	Additional guidance	Mark
(iii)	a description to include four from:		(4)
	take measurement without source (1)	measure/account for background (count)	
	place source in front of/near/close to detector (1)	DO NOT allow 'inside'	
	increase the distance (between source and detector) (1)	allow reverse argument by starting with detector long way away from source	
	measure distance (from source to detector) (1)		
	take reading from the screen/counter (1)		
	until reading gets to background value /constant value (1)	allow zero as constant value	
	use same time for each count (1)	mention of (count) rate	
	repeat / check when down to low values (1)		

Q3.

Question Number	Answer	Acceptable answers	Mark
	A description including any four from: (there are) 89 particles in the nucleus (1) protons (1) (there are) 36 (protons) (1) neutrons (1) (there are) 53 (neutrons) (1) i.e. 36 protons and 53 neutrons gains four marks	ignore all references to electrons (its) {mass/nucleon} number / RAM / A_r / A is 89 {atomic/proton} number / positive charge / Z = 36 Numbers must be correctly linked to gain credit e.g. 36 neutrons gets 1 mark (for neutrons) 53 protons and 36 neutrons gains two marks (for protons and neutrons) 89 protons and neutrons gets 3 marks (altogether there are) 89 protons and neutrons. 36 are protons gains 4 marks	**(4)**

Q4.

Question Number	Answer	Acceptable answers	Mark
(i)	A protons B neutrons C electrons	OR A neutrons B protons C electrons	**(3)**

Question Number	Answer	Acceptable answers	Mark
(ii)	**12**		**(1)**

Q5.

	Answer	Acceptable answers	Mark
(a)(i)	☒ **C** the same as the charge on the proton		**(1)**
(a)(ii)	☒ **A** electrons		**(1)**
(b)(i)	222	4 less/4fewer	**(1)**
(b)(ii)	86	2 less/2fewer	**(1)**
(c)	A description including **two** of: • Kill/damage cells(1) • affecting DNA (1) • (causing) mutation (1) • by ionisation (1) • make cell reproduce rapidly (1) • cause cancer (1) • (radiation) burns (1) • (radiation) sickness (1)		**(2)**
(d)	An explanation linking any suitable precaution to a sensible reason: Eg lead/shielding (1) (because it) stops/absorbs radiation (1) Use of radiation meters (1) measure/warn of radiation received (1) put up signs (1) (to)keep people away from radiation (1) increasing distance (1) (to)reduce intensity (1) to reduce dose(1) by limiting the number of X-rays taken (1)	source locked away for shielding (worker) leaves room inverse square law	**(2)**

Q6.

Question Number	Answer	Additional guidance	Mark
	substitution (1) $\frac{1.6726 \ (\times 10^{-27})}{9.1094 \ (\times 10^{-31})}$ evaluation (1) 1836 evaluation to 2 sf (1) 1800	Allow 1 mark for answers that round to 1.836 to any power of ten for this mark 1.836×10^3 OR 1.80×10^3 accept 1840 or any rounding of 1836.125 1.8×10^3 **any number** shown to 2 sf gets this mark award full marks for the correct answer without working	**(3)**

Q7.

Question number	Answer	Mark
	• evidence of division of activity by 2 (1) • 120 (Bq) (1)	**(2)**

Q8.

	Answer	Additional guidance	Mark
(i)	260 (g)		(1) AO2

	Answer	Additional guidance	Mark
(ii)	(54 days is) 3 half-lives (1) 65 (1)	260 ÷ 2 (÷ 2) or 520 ÷ 2 ÷ 2 (÷ 2) 18, 36, 54 (represents 3 half-lives) 54/18 = 3 (half-lives) ecf answer to 6ci ÷ 4 130 scores 1 mark award full marks for the correct answer without working	**(2)** **AO2**

Q9.

Question number	Answer	Additional guidance	Mark
	$\frac{45\,000}{5}$ (1) 9000 (kg) (1)	 award full marks for the correct answer without working	**(2)**

Q10.